Project Title : GreenMind: Deep Reinforcement Learning for Sustainable Buildings

TEAM : 13

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PROBLEM BACKGROUND

Buildings use a lot of energy. Heating, ventilation, air-conditioning, lighting and ventilation systems are the reasons for high energy use. Most buildings today use management systems. These systems follow set schedules or fixed rules to control how things work.. Real-world building conditions change all the time. Things like who is inside outdoor weather, indoor temperature, humidity, carbon dioxide levels and the time of day can all shift quickly. Because of this old ways of controlling systems often waste energy. For example they might keep heating or cooling when no one is around. They also don't adjust well when conditions change. As the need for energy efficiency grows buildings must become smarter. Intelligent control systems that learn from real-time data can help save energy. These systems respond better, to situations. They make sure energy is used when needed. This leads to sustainable buildings that work better and cost less to operate.

PROBLEM STATEMENT

This project seeks to build and test a model-free deep reinforcement learning policy. The goal is to find the way to manage energy use keep people comfortable indoors and maintain good air quality. The problem is a continuous-action Partially Markov Decision Process or POMDP. Here an agent uses real-time sensor readings—like temperature, humidity, CO_2 levels how many people are inside and outdoor weather—to send continuous commands. These commands control dampers, fans and chilled-water control valves. The policy is designed to balance goals. It tries to

use as power as possible while keeping the indoor environment comfortable and keeping CO_2 concentration as low as possible measured in parts, per million.

WHY REINFORCEMENT LEARNING?

Adaptability to Dynamic Environments means that buildings are constantly changing because of factors such as how many people are inside the temperature and the time of day. Traditional controllers set fixed limits, which often waste energy when rooms are empty. An RL agent on the hand changes its actions carefully by watching the environment in real time.

Multi-Objective Optimization means working on goals at once instead of just one. A model-free deep RL policy can understand a reward rule and learn to use less energy while still keeping the room comfortable and the air clean.

Managing Complex Continuous Actions means that running HVAC systems is a job, like a Partially Observable Markov Decision Process, with continuous moves. Deep RL helps because it can read sensor data and turn it into smooth action commands.

LITERATURE REVIEW

S No	Paper	Year	Source
1	Hybrid Reinforcement Learning for Occupant-Centric Building Control: A Review and Deployment Framework for Co-Optimizing Energy, Comfort, and Indoor Air Quality	2026	https://www.sciencedirect.com/science/article/pii/S0306261926000449?via%3Dihub
2	Occupancy-Driven HVAC Control Optimization via LSTM and Deep Reinforcement Learning for Enhanced Indoor Air Quality, Thermal Comfort and Energy Efficiency	2025	https://www.sciencedirect.com/science/article/abs/pii/S0360132325009746?via%3Dihub
3	Occupant-Centric HVAC and Window Control: A Reinforcement Learning Model for Enhancing Indoor Thermal Comfort and Energy Efficiency	2024	https://www.sciencedirect.com/science/article/abs/pii/S0360132324000398?via%3Dihub

4	A Practical Deep Reinforcement Learning Framework for Multivariate Occupant-Centric Control in Buildings	2022	https://www.sciencedirect.com/science/article/abs/pii/S0306261922010297?via%3Dihub
5	Towards Self-Learning Control of HVAC Systems with the Consideration of Dynamic Occupancy Patterns: Application of Model-Free Deep Reinforcement Learning	2022	https://www.sciencedirect.com/science/article/abs/pii/S0360132322009775
6	Intelligent Multi-Zone Residential HVAC Control Strategy Based on Deep Reinforcement Learning	2021	https://www.sciencedirect.com/science/article/abs/pii/S030626192031535X?via%3Dihub

EXISTING SYSTEMS / PRODUCTS STUDY

I see that Household energy management research is moving more toward Deep Reinforcement Learning (DRL) to make climate control in homes focused on people especially for heating, ventilation and air-conditioning (HVAC) systems. I note that of using fixed setpoints that are hard coded modern DRL models widen the set of data they look at to include many real-time measurements, such as how many people are in a room the indoor dry-bulb temperature, the humidity, the carbon-dioxide level, in each zone weather forecasts and the time of day.

Parameter	**Current Research**
Control Domain	DRL is being used more and more for heating, ventilation and air conditioning control. DRL helps make HVAC control focus on people to manage zones at the same time and to keep the system running all the time.
Observation Vectors	Studies use combinations of occupancy levels, indoor temperature, humidity, carbon dioxide levels, outdoor conditions, weather details and information about the people inside. Studies look at occupancy levels. Also look at indoor temperature. Studies look at humidity and carbon dioxide levels. Studies consider conditions and weather

	details. Studies take into account information, about the people.
Occupancy Handling	Recent research often adds occupancy and it also adds occupancy prediction using methods such, as LSTM and occupant-centric models.
Optimization Goals	Existing studies mainly focus on reducing energy use and improving comfort. More recent research is starting to include air quality, especially carbon dioxide levels as an important factor.
Action Space	DQN and DDQN studies often use HVAC actions. DDPG and other DRL methods provide control. DQN and DDQN studies often use HVAC actions. DDPG and other DRL methods provide control.
Multi-Zone Control	Multi-agent and multi-zone DRL approaches have been introduced to manage heating, ventilation and air conditioning systems across parts of a building. These methods help coordinate how HVAC systems work in each zone. The goal is to improve energy efficiency and comfort by allowing each zone to operate based on its needs. By using agents each zone can make decisions independently while still working together. This setup supports control, over temperature and air quality in large buildings.
Training Environment	Most studies rely on building simulation environments for training and evaluation. Other studies progress toward real-building deployment.

Safety During Training	RL agents learn through trial and error and studies have identified that RL agents may experience thermal conditions during learning.
Deployment	A few studies show deployment, in real buildings. Still I hear many reviews say that deployment is a challenge.

GAP ANALYSIS

- Limited integration of objectives. I notice that existing studies mainly focus on energy consumption and thermal comfort. Recent studies have started to add IAQ. A unified framework that optimizes energy, thermal comfort, IAQ and safety at once remains a challenge.

- Discrete versus control. I see that several studies use DQN or DDQN for actions. Continuous-control methods such as DDPG have also been tried. Yet integrating HVAC control with IAQ, occupancy and safety constraints is still an open challenge.

- Dynamic occupancy uncertainty. I see that occupancy data is often used in HVAC control even with LSTM-based prediction. Still changing and uncertain occupancy patterns can hurt the effectiveness and generalization of the learned control policy.

- Safety during RL training. I see that RL learns by trial and error which can create temperature or IAQ conditions while exploring. Existing studies point out this problem. Still putting safety violations into the learning objective is a major challenge.

- Energy–comfort–IAQ trade-off. I see that more ventilation can lower CO_2 and improve IAQ but it can raise HVAC energy consumption. Cutting HVAC operation saves energy. Can hurt comfort or air quality. Current approaches do not fully solve this trade-off.

- Simulation-to-real-world gap. I see that most RL-HVAC research uses building simulators to train safely and practically. Some studies show real-building deployment. Still moving a simulation-trained policy, to shifting real-world conditions remains difficult.

PROPOSED RESEARCH DIRECTION

- First train the RL agent inside building simulators such as EnergyPlus. This lets the RL agent learn how buildings heat and cool safely without touching equipment.

- Start by exploring DQN and Double DQN. These algorithms will help the RL agent handle HVAC controls and set a baseline for performance.

- Next extend the approach to continuous-action algorithms like PPO and SAC. This will let the RL agent control fan speeds, dampers and chilled-water valves smoothly.

- Evaluate the learned policy by comparing it with rule-based HVAC controls, for example ASHRAE-based control. Use metrics, like energy consumption, thermal comfort and IAQ.

- This study focuses on developing a reward function. The reward will balance energy efficiency, occupant comfort, indoor air quality and safety.